



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201- ENGINEERING MATHEMATICS II

UNIT V & COMPLEX FUNCTIONS

3.MAPPING OF COMPLEX FUNCTIONS

3.1 CONFORMAL MAPPING

3.2 TYPES OF TRANSFORMATIONS & PROBLEMS

3.3 CRITICAL POINTS & PROBLEMS

3.4 PRACTICE PROBLEMS

Introduction:

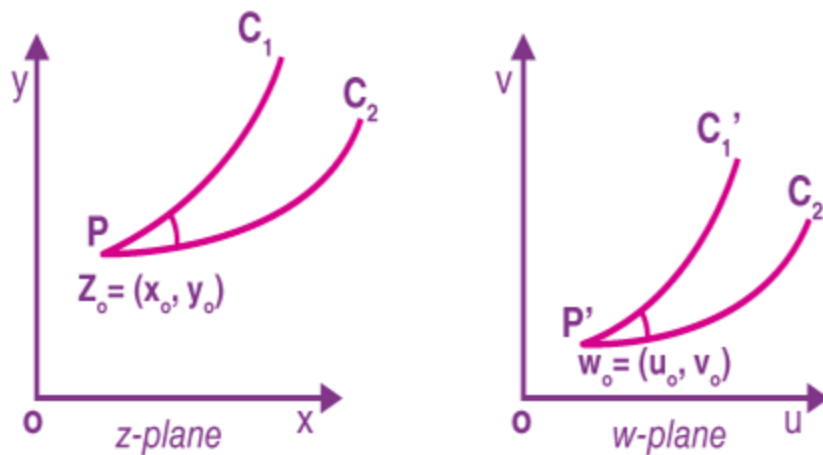
3.1 Conformal mapping is a function defined on the complex plane which transforms a given curve or points on a plane, preserving each angle of that curve. If $f(z)$ is a complex function defined for all z in C , and $w = f(z)$, then f is known as a transformation which transforms the point $z = x + iy$ in z -plane to $w = u + iv$ in w -plane. If this transformation preserves the angles between curves in both magnitude and sense (clockwise or counterclockwise), then the mapping is called conformal mappings.

If there is a transformation such that the magnitude of the angles between the curves is identical, but the sense (clockwise or counterclockwise) is opposite. In that case, the transformation is called **isogonal**.

Definition of Conformal Mapping

Consider the transformations $u = u(x, y)$ and $v = v(x, y)$, which maps a point $P(x_0, y_0)$ in the z -plane to a point $P'(u_0, v_0)$ in the w -plane. Let curves C_1 and C_2 intersect at point $z_0 = (x_0, y_0)$ in z -plane is mapped into curves C_1' and C_2' in w -plane intersecting at $w_0 = (u_0, v_0)$.

If the transformation is such that the angle between C_1 and C_2 at z_0 is equal both in magnitude and sense to the angle between the curve C_1' and C_2' at w_0 , it is known as conformal mapping at $z_0 = (x_0, y_0)$.



Note: i) The scaling factor and the rotation angle for a given transformation $f(z)$ depend only on z and not on curves passing through z .

ii) Conformality depends on the given point. It is different at various points in the plane.

Conditions for Conformal Mapping

The sufficient condition for a transformation $w = f(z)$ to be a conformal mapping is:

Let $f(z)$ be an analytic function of z in a domain D of the z -plane and let $f'(z) \neq 0$ inside D . Then the mapping $w = f(z)$ is conformal at all points of D .

The necessary condition for a transformation being a conformal mapping is:

If $w = f(z)$ represents a conformal transformation of a domain D of z -plane into a domain D' of w -plane, then $f(z)$ is an analytic function of z in D' .

3.2 TRANSFORMATIONS - Types of Transformations

The geometric **transformation** is a bijection of a set that has a geometric structure by itself or another set. If a shape is transformed, its appearance is changed. After that, the shape could be congruent or similar to its preimage. The actual meaning of transformations is a change of appearance of something.

After all the transformations, these shapes still has the same size, angles, lengths and area.

Types of Transformations

When the size of a shape is increased or reduced then the image of the shape will be similar to the pre-image. The [similar figures](#) have dimensions equal in proportion.

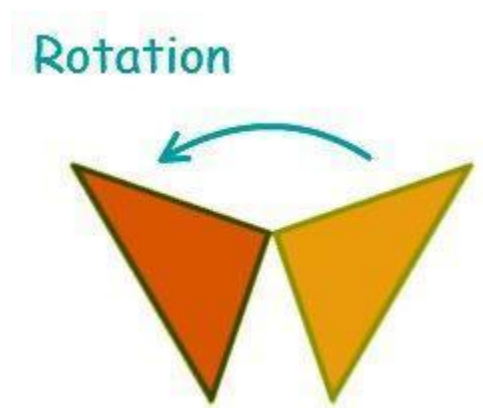
But in the case of congruent, the transformation of objects is done by using rotation, reflection or translation. The shape is turned or flipped to transform into another shape.

There are three major types of transformations namely:

- Rotation
- Translation
- Dilation (Magnification)

Rotation

This type of transformation has an object about a fixed point without changing its size or shape.

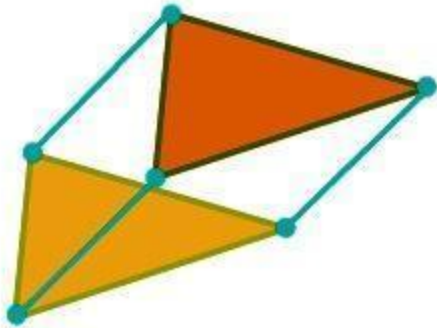


In the above figure, you can see, that the shape is rotated to form its image.
Learn more about [rotation](#) here.

Translation

This type of translation is defined as moving the object in space by keeping its size, shape or orientation constant. In a translation, each point of the shape must be moved in the same direction and for the same distance. When you are doing a translation, the primary object is called the pre-image, and the object after the translation is called the image.

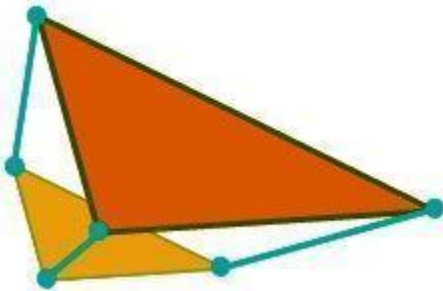
Translation



Dilation (Magnification)

This type of translation expands or contracts the object by keeping its orientation or shape the same. This is also known as resizing.

Resizing



Some standard transformations

Translation:

The transformation $w = C + z$, where C is a complex constant, represents a translation.

Let $z = x + iy$, $w = u + iv$ and $C = a + ib$

Given $w = z + C$,

$$(i.e.) u + iv = x + iy + a + ib$$

$$\Rightarrow u + iv = (x + a) + i(y + b)$$

Equating the real and imaginary parts, we get $u = x + a$, $v = y + b$ Hence the image of any point

$p(x, y)$ in the z -plane is mapped onto the point $p'(x + a, y + b)$ in the w -plane.

Similarly, every point in the z -plane is mapped onto the w plane.

If we assume that the w -plane is super imposed on the z -plane, we observe that the point (x, y) and hence any figure is shifted by a distance $|C| = \sqrt{a^2 + b^2}$ in the direction of C i.e., translated by the vector representing C .

Hence this transformation transforms a circle into an equal circle. Also the corresponding regions in the z and w planes will have the same shape, size and orientation.

Example: What is the region of the w plane into which the rectangular region in the Z plane bounded by the lines $x = 0$, $y = 0$, $x = 1$ and $y = 2$ is mapped under the transformation $w = z + (2 - i)$

Solution: Given $w = z + (2 - i)$ (i.e.)

$$u + iv = x + iy + (2 - i) = (x + 2) + i(y - 1)$$

Equating the real and imaginary parts

$$u = x + 2, v = y - 1$$

Given boundary lines are

$$x = 0$$

$$y = 0$$

$$x = 1$$

$$y = 2$$

transformed boundary lines are

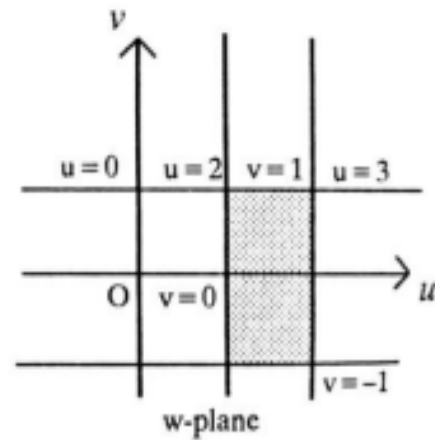
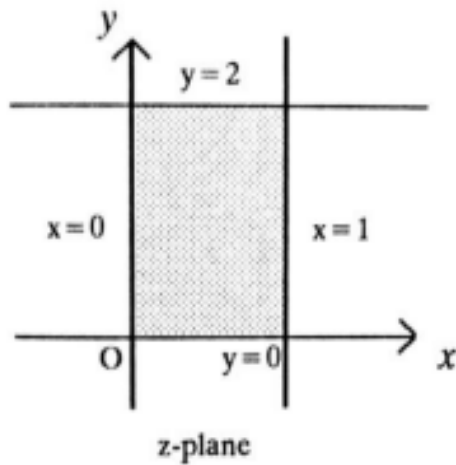
$$u = 0 + 2 = 2$$

$$v = 0 - 1 = -1$$

$$u = 1 + 2 = 3$$

$$v = 2 - 1 = 1$$

Hence, the lines $x = 0$, $y = 0$, $x = 1$, and $y = 2$ are mapped into the lines $u = 2$, $v = -1$, $u = 3$, and $v = 1$ respectively which form a rectangle in the w plane.



Example: Find the image of the circle $|z| = 1$ by the transformation $w = z + 2 + 4i$

Solution: Given $w = z + 2 + 4i$

(i.e.) $u + iv = x + iy + 2 + 4i$

$$= (x + 2) + i(y + 4)$$

Equating the real and imaginary parts,

we get $u = x + 2$, $v = y + 4$,

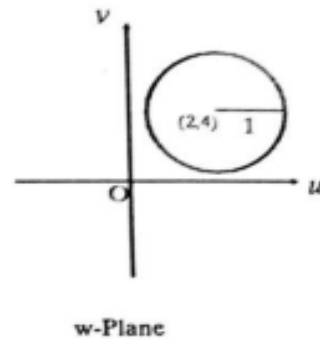
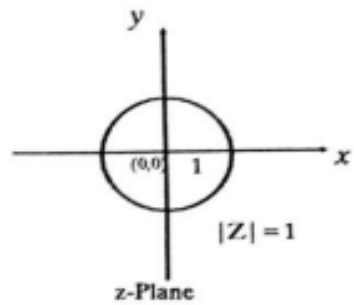
$$x = u - 2, \quad y = v - 4,$$

Given $|z| = 1$

$$(i.e.) \quad x^2 + y^2 = 1$$

$$(u - 2)^2 + (v - 4)^2 = 1$$

Hence, the circle $x^2 + y^2 = 1$ is mapped into $(u - 2)^2 + (v - 4)^2 = 1$ in w plane which is also a circle with centre $(2, 4)$ and radius 1



Magnification and Rotation

The transformation $w = cz$, where c is a complex constant, represents both magnification and rotation.

This means that the magnitude of the vector representing z is magnified by $a = |c|$ and its direction is rotated through angle $\alpha = \text{amp}(c)$. Hence the transformation consists of a magnification and a rotation.

Example: Determine the region 'D' of the w -plane into which the triangular region D enclosed by the lines $x = 0$, $y = 0$, $x + y = 1$ is transformed under the transformation $w = 2z$.

Solution:

$$\text{Let } w = u + iv, \quad z = x + iy$$

$$\text{Given } w = 2z$$

$$u + iv = 2(x + iy)$$

$$u + iv = 2x + i2y$$

$$u = 2x \quad \Rightarrow x = u/2,$$

$$v = 2y \quad \Rightarrow y = v/2$$

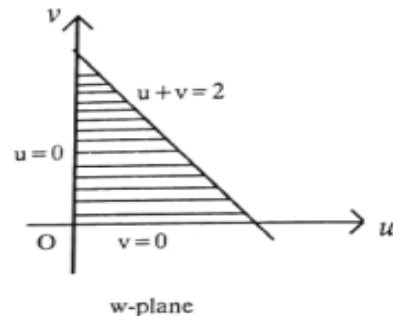
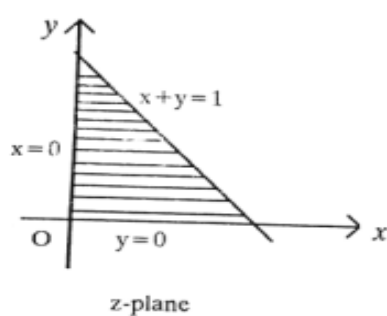
Given region (D) whose boundary lines are		Given region (D) whose boundary lines are
$x = 0$	$\Rightarrow$	$u = 0$

In z	$y = 0$	$\Rightarrow$	$v = 0$	the
	$x + y = 1$		$\frac{u}{2} + \frac{v}{2} = 1 [\because x = \frac{u}{2}, y = \frac{v}{2}]$ $(i.e.) u + v = 2$	

plane the line $x = 0$ is transformed into $u = 0$, in the w plane.

In the z plane the line $y = 0$ is transformed into $v = 0$, in the w plane.

In the z plane the line $x + y = 1$ is transformed into $u + v = 2$ in the w plane.



Example:

Find the image of the circle $|z| = \lambda$ under the transformation $w = 5z$.

Solution: Given $w = 5z$

$$|w| = 5|z|$$

$$i.e., |w| = 5\lambda \quad [\because |z| = \lambda]$$

Hence, the image of $|z| = \lambda$ in the z plane is transformed into $|w| = 5\lambda$ in the w plane under the transformation $w = 5z$.

Example: Find the image of the circle $|z| = 3$ under the transformation $w = 2z$

Solution:

Given $w = 2z, \quad |z| = 3$

$$|w| = (2)|z|$$

$$= (2)(3),$$

Since $|z| = 3 = 6$ Hence, the image of $|z| = 3$ in the z plane is transformed into $|w| = 6$ in the w plane under the transformation $w = 2z$.

Example: Find the image of the region $y > 1$ under the transformation

$$w = (1 - i)z.$$

Solution:

$$\text{Given } w = (1 - i)z.$$

$$u + v = (1 - i)(x + iy)$$

$$= x + iy - ix + y$$

$$= (x + y) + i(y - x)$$

$$\text{i. e. , } u = x + y, \quad v = y - x$$

$$u + v = 2y$$

$$u - v = 2x$$

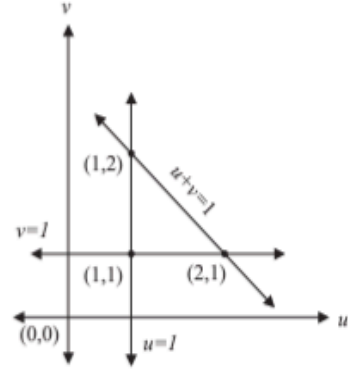
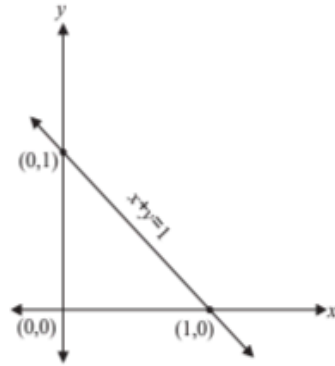
$$y = (u+v)/2, \quad x = (u-v)/2$$

Hence, image region $y > 1$ is $(u+v)/2 > 1$ i.e., $u + v > 2$ in the w plane.

Example:

Consider the transformation $w = z + (1 + i)$ and determine the region in the w -plane corresponding to the triangular region bounded by the lines $x = 0, y = 0$ and $x + y = 1$ in the z -plane.

Solution: The given triangular region bounded by the lines $x = 0, y = 0$ and $x + y = 1$ is shown in fig. Then the vertices of the triangular region are $(0, 0), (1, 0)$ and $(0, 1)$. Now, the given transformation is



$$\begin{aligned}
 w &= z + (1 + i) \\
 \Rightarrow u + iv &= (x + iy) + (1 + i) \\
 \Rightarrow u + iv &= (x + 1) + i(y + 1) \\
 \Rightarrow u &= x + 1 \text{ and } v = y + 1.
 \end{aligned}$$

$$\begin{aligned}
 x = 0 &\Rightarrow u = 1; & y = 0 &\Rightarrow v = 1; & x + y = 1 &\Rightarrow u - 1 + v - 1 = 1 \\
 & & & & &\Rightarrow u + v = 3
 \end{aligned}$$

The line $x = 0$ maps into $u = 1$, which is also the vertical line in w -plane.

Also, the line $y = 0$ maps into $v = 1$, which is the horizontal line in w -plane.

And, the line $x + y = 1$ maps into the line $u + v = 3$.

Hence, the region becomes triangle bounded by the lines $u = 1, v = 1, u + v = 3$;

Example Find the image of the circle $|z| = 2$ under the transformation $w = iz + 1$.

Solution: Let $w = u + iv$ and $z = x + iy$. Then

$$\begin{aligned}w &= iz + 1 \\ \Rightarrow u + iv &= i(x + iy) + 1 = ix - y + 1 = (1 - y) + ix\end{aligned}$$

$$\Rightarrow u = 1 - y \text{ and } v = x$$

$$\Rightarrow y = 1 - u \text{ and } x = v$$

Now,

$$\begin{aligned}|z| = 2 &\Rightarrow |z|^2 = 4 \\ &\Rightarrow |x + iy|^2 = 4 \\ &\Rightarrow x^2 + y^2 = 4 \\ &\Rightarrow v^2 + (1 - u)^2 = 4 \\ &\Rightarrow (u - 1)^2 + v^2 = 4\end{aligned}$$

which is equation of the circle centered at $(1, 0)$ and radius is 2. Thus the transformation rotates the circle by $\frac{\pi}{2}$ and translate its by unity to the right. $\square$

3.3 CRITICAL POINTS & PROBLEMS

Critical Point

Let $f(z)$ be a nonconstant analytic function in a domain D . If $f'(z_0) = 0$ for some $z_0 \in D$, then z_0 is called the critical point of the transformation. Clearly, $z = 0$ is the critical point of the transformation $w = f(z) = 2z^3 - 1$. It is also to be noted that constant map has no critical points.

Example: Find the critical points of the transformation $w^2 = (z - \alpha)(z - \beta)$.

Solution:

$$\text{Given } w^2 = (z - \alpha)(z - \beta) \dots (1)$$

Critical points occur at $dw/dz = 0$ and $dz/dw = 0$

Differentiation of (1) w. r. to z , we get

$$\begin{aligned} \Rightarrow 2w \, dw/dz &= (z - \alpha) + (z - \beta) \\ &= 2z - (\alpha + \beta) \end{aligned}$$

$$\Rightarrow dw/dz = 2z - (\alpha + \beta) / 2w \dots (2)$$

Case (i) $dw/dz = 0$

$$\begin{aligned} \Rightarrow 2z - (\alpha + \beta) / 2w &= 0 \\ \Rightarrow 2z - (\alpha + \beta) &= 0 \\ \Rightarrow 2z &= \alpha + \beta \\ \Rightarrow z &= \alpha + \beta / 2 \end{aligned}$$

Case (ii) $dz/dw = 0$

$$\begin{aligned} \Rightarrow 2w / 2z - (\alpha + \beta) &= 0 \\ \Rightarrow w / z - \alpha + \beta / 2 &= 0 \\ \Rightarrow w &= 0 \\ \Rightarrow (z - \alpha)(z - \beta) &= 0 \\ \Rightarrow z &= \alpha, \beta \end{aligned}$$

$\therefore$ The critical points are $\alpha + \beta/2$, α and β .

Example: Find the critical points of the transformation $w = z^2 + 1/z^2$.

Solution:

$$\text{Given } w = z^2 + 1/z^2 \dots (1)$$

$$\text{Critical points occur at } dw/dz = 0 \text{ and } dz/dw = 0$$

Differentiation of (1) w. r. to z , we get

$$\begin{aligned} \Rightarrow dw/dz &= 2z - 2/z^3 \\ &= 2z^4 - 2/z^3 \end{aligned}$$

$$\text{Case (i) } dw/dz = 0$$

$$\Rightarrow 2z^4 - 2/z^3 = 0$$

$$\Rightarrow 2z^4 - 2 = 0$$

$$\Rightarrow z^4 - 1 = 0$$

$$\Rightarrow z = \pm 1, \pm i$$

$$\text{Case (ii) } dz/dw = 0$$

$$\Rightarrow z^3 / (2z^4 - 2) = 0$$

$$\Rightarrow z^3 = 0 \Rightarrow z = 0 \therefore \text{The critical points are } 1, \pm i, 0$$

Example: Find the critical points of the mapping $w = e^{2z} - 2iz + 3$.

Solution:

Since $w = f(z) = e^{2z} - 2iz + 3$ is non-constant and analytic for all $z \in \mathbb{C}$, it follows that its critical points are the values of z such that $f'(z_0) = 0$.

Hence $2e^{2z} - 2i = 0$

i.e. $e^{2z} = i = e^{i(2n\pi + \pi/2)}$

i.e. $z = i(n\pi + \pi/4), \quad n \in \mathbb{I}.$

The critical points of the transformation $w = (z - 1)(z - 2)$

A) $z = 1$

B) $z = 2$

C) $z = \frac{3}{2}$

D) $z = \frac{4}{5}$

Correct Answer:

C) $z = \frac{3}{2}$

Description for Correct answer:

$$w = z^2 - 3z + 2$$

$$\frac{dw}{dz} = 2z - 3$$

Critical points is given by,

$$\frac{dw}{dz} = 0$$

$$\Rightarrow 2z - 3 = 0 \Rightarrow z = \frac{3}{2}$$

3.4 PRACTICE PROBLEMS

1. Find the image of the circle $|z| = a$ under the following transformation

a) $w = z + 2 + 3i$ $Ans : (u - 2)^2 + (v - 3)^2 = a^2$

b) $w = 2z$ $Ans : |w| = 2a$

c) $w = (1 + i)z + 2 - i$ $Ans : (u - 2)^2 + (v + 1)^2 = 2a^2$

2. Find the image of the circle $|z| = 2$ by the transformation

Find the image of the circle $|z| = a$ under the following transformation

- a) $w = z + 3 + 2i$ $Ans : (u-3)^2 + (v-2)^2 = 2^2$
- b) $w = 3z$ $Ans : u^2 + v^2 = 36$
- c) $w = (1 + 2i)z + (3 + 4i)$ $Ans : (u-3)^2 + (v-4)^2 = 20$

2. Find the critical points of the transformation.

$$w = \frac{1}{2} \left(z e^{-\alpha} + \frac{e^{\alpha}}{z} \right) \quad Ans : z = \pm e^{\alpha}$$

DISCUSSION QUESTIONS:

1. What is special about a conformal map?
2. What is the necessary condition for a conformal mapping ?
3. What is the other name for conformal mappings?